



Analysis of Visual Patterns Through the EEG Signal: Color Study

Ana Rita Teixeira^{1,2}  and Anabela Gomes^{3,4} 

¹ Polytechnic Institute of Coimbra- ESEC, UNICID, Coimbra, Portugal
ateixeira@ua.pt

² Institute of Electronics and Informatics Engineering of Aveiro, University of Aveiro, Aveiro, Portugal

³ Polytechnic Institute of Coimbra- ISEC, Coimbra, Portugal

⁴ CISUC—Department of Informatics Engineering, University of Coimbra, Coimbra, Portugal

Abstract. This study quantifies the amplitude and latency variability of the visual pattern evoked potential (P100) using the MUSE device and compares this variability between two populations: color blind versus normal eyes.

The main objective of this work is to identify the impact of color on brain waves in color-blind and non-color-blind individuals. Thus, P100 wave was used and the amplitude and latency generated by each visual stimulus was analyzed. The results show that different colors present different amplitudes and latencies in the two samples and between colors. The Wilcoxon test was used to compare the samples. In both samples, blue and red color with 100% saturation present higher amplitudes. Blue amplitudes value is statistically different considering the two samples. The blue amplitude is higher in color-blind individuals, meaning that it causes greater stimulation in brain activity. On the other hand, although the amplitude of the red color is high, it is similar in both samples. Yellow, Blue, Yellow and Saturated Green present discrepant values of P100 wave amplitude in both samples. Amplitudes are much higher in non-color-blind individuals than in color-blind individuals. According to this study, different colors can cause different amplitudes and latency in P100 waves, considering color blind and non-color-blind people. In addition, our findings also confirm previous studies that indicate that energy in theta and alpha bands can be used to discriminate some colors. In summary, this study is useful to find metrics to classify colors helping in marketing strategies and design typography. The P100 wave used in this kind of strategies is a novelty in the literature and can be considered for future studies.

1 Introduction

Nowadays, with the accelerated development of technology and the long exposure of the population to the screens, color and its use has begun to play a significant role. Color perception is one of the most important cognitive features of human brain. Different colors lead to different cognitive activities and different mental arousal levels as revealed by power spectral density obtained from EEG signals [1]. There are several studies in the literature regarding color in different research areas, such as psychology [2, 3], neuromarketing [1, 3–6] and design [7, 8]. Color has aesthetic value, and it also

influences cognition, emotion, and behavior. Cool colors (green, blue, and violet) are associated with being comfortable, relaxing, peaceful, and calming, thus reducing stress and anxiety levels. On the other hand, warm colors (red, yellow, and orange) are more arousing, which can stimulate human feelings and activate people. Neutral colors have less emotional content and, thus, lesser psychological impact [2]. Studies have revealed that colored multimedia learning materials have a positive effect on learner's emotion and learning where it was assessed by subjective/objective measurements [3]. Color has an important role in object recognition and visual memory [9]. Decoding color in the human brain is helpful to understand the mechanism of visual cognitive process and evaluate memory ability as well as to understand the impact that each color has in the human brain. Colors define our emotions and many times target our impulses and decisions. Therefore, picking the right colors will have a dramatic effect on the engagement results. Human feelings and moods are changeable overtime thus, every marketing manager should know the importance of colors and make their products or packaging accordingly [4]. The color is important in marketing and packaging industry, so, neuro-marketing research, based on color stimuli is considered to be an important tool for market research. Brain-Computer Interaction (BCI) technology can be used in several areas and has recently gained increased interest with diverse applications in the area of Human Computer Interaction (HCI). The color is one of the central elements in perceiving information that, when properly used, can provide better readability and understanding of the information to be communicated. In this sense, this multidisciplinary work (Theory of Visual Perception, HCI and cognitive neuroscience) examines how the brain processes color information using EEG technology. So, BCI technology is used in neuromarketing to study the brain's responses to marketing stimuli [10].

This paper is organized as follows: First, the introduction of the problem is done; second, the methodology (samples, protocol and devices) is presented, after which we have the results, and finally we present the discussions and some conclusions.

2 Methodology

2.1 Sample

The current study included 25 people, aged between 18 and 60 years, 12 women and 13 men. Of the 25 participants, 5 are people who have some type of color blindness (2 have high-grade deuteranopia color blindness, 1 has color blindness of the same type at a moderate degree, and 2 have tritanopia-type color blindness in a moderate degree) and the remaining 20 do not have any degree of deficiency in color distinction.

2.2 Protocol

The experimental protocol based on [12], consisted of viewing 10 distinct colors on the screen of a PC interspersed with the gray color. Initially, the colors of the RGB model with 100% saturation (S.) were presented. Subsequently, the same shades of color were presented, with 50% saturation. The duration time was 2 s in gray and only 1 s in the remaining colors, making a total of 21 colors and 32 s of viewing. The protocol starts

with grey color, then the other colors and between them the grey color is shown for 2 s and the protocol finishes with the grey color. In the following table 1 it is possible to check the order of the colors used between grey color:

Table 1. Designation Color, where S. means Saturated

# Color	RGB	Name	# Color	RGB	Name
2	(0; 0; 255)	S. Blue	12	(62; 62; 194)	Blue
4	(255; 255; 0)	S. Yellow	14	(193; 194; 62)	Yellow
6	(255; 0; 255)	S. Violet	16	(218; 150; 218)	Violet
8	(255; 0; 0)	S. Red	18	(194; 62; 62)	Red
10	(0; 255; 0)	S. Green	20	(62; 194; 62)	Green

2.3 Device

To carry out this project the BCI device used is an EEG acquisition equipment, in the form of a headband, called InteraXon’s MUSE 2 (InteraXon, 2018) designed to be placed on the front of the head, being very practical, lightweight and adjustable [11]. With a sampling of 256 Hz, the EGG system consists of 5 dry sensors: 2 on the forehead (F7 and AF8), 2 behind the ears (TP9 And TP10) and a reference sensor for calibration that is in the center of the forehead (FPZ) (Muse), Fig. 1.



Fig. 1. Muse and Electrodes placement

3 Results

The main goal of this work is to identify the impact of color on brain waves in colorblind and non-color-blind individuals. Thus, the data examined in this study refers to acquisitions of the P100 wave (Fig. 2), namely, amplitude and latency, generated by each visual stimulus.

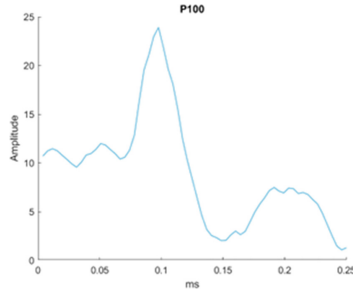


Fig. 2. P100 wave

Aspect of selective attention such as task difficulty will affect the latency and amplitude of the P100. For example, if a task is more difficult, it will take longer for the P100 to occur and the amplitude will be higher. On the other hand, if the task is easier, it will take less time for the P100 to occur and the amplitude will be lower. Stimulus encoding also affects latency and amplitude of the P100. If the stimulus is more novel or requires more effort to encode, it will take longer for the P100 to occur and the amplitude will be higher. If the stimulus is more familiar or requires less effort to encode, it will take less time for the P100 to occur and the amplitude will be lower. This effort can be seen in the P100 wave, which is a component of the ERP (event-related potential) that is generated in response to a stimulus. This wave is typically seen in the range of 80–120 ms and is thought to represent the selection and allocation of attentional resources. Therefore, increased task difficulty may lead to increased latency and amplitude of the P100 wave.

3.1 Amplitude

In Fig. 3, the P100 wave amplitudes are represented considering the color-blind individuals and non-color-blind individuals.

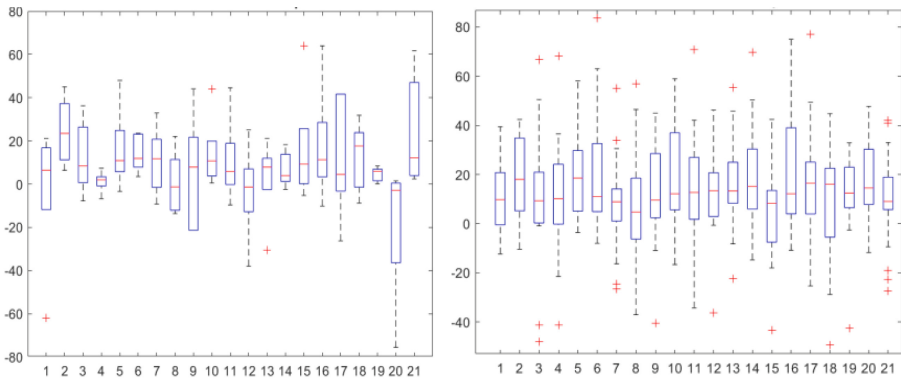


Fig. 3. P100 wave amplitudes for color-blind individuals (left boxplot) and for non-color-blind individuals (right boxplot) considering each color.

In both samples it is irrefutable that the most impacting color visually and which promotes significantly higher amplitude values is the blue color, with 100% saturation (Table 2). This amplitude value is slightly higher in color-blind individuals than in non-colored-blind individuals, meaning this causes greater stimulation in brain waves and can influence any decision-making. Then, in both samples the color 18 (Red) is the color with the highest impact on brain waves, and this value does not differ much in the two Samples. Color 6 (S. Violet), color 16 (Violet) and color 10 (S. Green) resulted in intermediate values, i. e., not in P100 wave high peaks, but were still indifferent, giving rise to values identical to each other and in both samples, as can be seen in the Table 2.

Table 2. Average values of the P100 wave amplitude along the color segment in the two Samples.

#	Color-Blind	Non-Color-Blind
2	23,49	18,03
4	1,98	10,11
5	11,86	11,08
8	-1,31	4,74
10	10,8	12,18
12	-1,4	13,47
14	3,96	15,17
16	11,29	12,19
18	17,63	16,05
20	-2,87	14,5

On the other hand, color 4 (S. Yellow), color 12 (Blue), color 14 (Yellow) and color 20 (Green), resulted in discrepant amplitude values of P100 wave in both samples. These figures are much higher in non-color-blind individuals than in color-blind individuals, demonstrating that these colors, effectively have the greatest impact on the brainwaves of the participants without any type of visual anomaly compared to the color - blind participants. There are in fact colors that did not have any effect in color blind participants, such as color 8 (S. Red), color 12 (Blue) and color 20 (Green), since the wave values are negative. Lastly the color which impacts both samples is color 8 (S. Red). In the sample of Color-blind Individuals, as already mentioned, it obtained negative effects and in the sample of non-color-blind individuals it was the color with the lowest amplitude among all.

3.2 Latency

In Fig. 4 is represented the latency color values for all colors along the experimental protocol.

Since latency is directly related to the response generated by the stimulus, the higher the latency value, the more amplitude occurs in the process of responding to the stimulus.

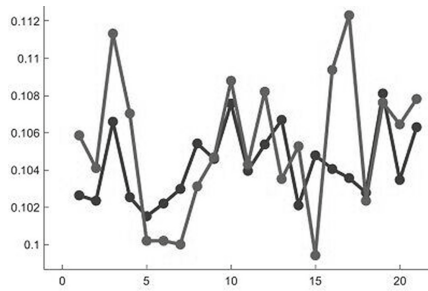


Fig. 4. Latency comparison between color-blind (grey) and non-color-blind (black) participants, considering the protocol colors (1–21)

In this sense, comparing the P100 wave latencies originated by each visual stimulus in the two samples, it can be concluded that there are no significant differences. It is, however, possible to highlight in non-color-blind individuals color 6 (S. Violet) with higher latency value compared to color-blind participants. On the other hand, color-blind participants obtained higher latency values in color 16 (Violet) and in color 20 (Green), these being the colors with the longest response time compared to the other sample.

3.3 Statistical Analysis

The Mann-Whitney U-test was used to compare the latency and amplitude between colors in the two groups of data, and the results showed that there was a statistically significant difference between some colors. This test is suitable for comparing two independent populations when the data is not normally distributed or when the variances of the two groups are not equal.

In this statistical analysis this test was chosen since only a small sample is available and the variable does not present a normal variation. In this sense, the hypotheses for the nonparametric test are presented as follows:

H0: No differences between colors

H1: Differences between colors

If the p-value obtained is lower than the alpha level of 0.05, which indicates that we can reject the null hypothesis (H0) and conclude that there are statistically significant differences between the two groups.

Color Amplitude in Color-Blind Individuals

In order to statistically analyze the amplitude of colors in the sample of color-blind individuals, it is immediately perceived, by the analysis of Fig. 1, that color 2 (S. Blue) is the one that has a more accentuated difference ($p < 0.05$) in relation to the others. Displays statistical differences with the remaining colors, except with color 16 (Violet) ($p = 0.076$). In addition, color 12 (Blue) is also incompatible with several colors, specifically, color 2 (S. Blue) ($p = 0.0006$), color 6 (S. Violet) ($p = 0.0294$) and color 18 (Red) ($p = 0.0154$). Color 6 (S. Violet) is statistically different with color 8 (S. Red) ($p = 0.033$), color 12 (Blue) ($p = 0.0294$) and color 20 (Green) ($p = 0.0083$). On the other hand,

there are statistically compatible colors, such as color 4 (S. Yellow), color 10 (S. Green) and color 16 (Violet), meaning that no difference is present.

Color Latency in Color-Blind Individuals

The statistical differences in the sample of color-blind individuals regarding latency is not so evident, highlighting color 16 (Violet) that differs statistically from color 10 (S. Green) ($p = 0.0024$), color 12 (Blue) ($p = 0.0032$) and color 18 (Red) ($p = 0.0012$).

Color Amplitude in Non-Color-Blind Individuals

Statistically, the differences between colors in amplitude between non-color-blind individuals are not accentuated, and the greatest differences are in saturated colors. Color 4 (S. Yellow) differs with more colors such as: color 14 (Yellow) ($p = 0.0147$), color 16 (Violet) ($p = 0.00226$), color 18 (Red) ($p = 0.0027$) and color 20 (Green) ($p = 0.0055$).

In addition, color 2 (S. Blue) differs with color 4 (S. Yellow) ($p = 0.0017$) and color 12 (Blue) ($p = 0.041$), as well as color 18 (Red) differs from color 6 (S. Violet) ($p = 0.041$) and color 12 (Blue) ($p = 0.038$). It should be noted that colors such as color 8 (S. Red) and color 10 (S. Green) are statistically compatible with all color tones.

Color Latency in Non-Color-Blind Individuals

In the statistical study of latency between colors, in non-color-blind individuals, it is evident that only color 10 (S. Green) is statistically incompatible with: color 2 (S. Blue) ($p = 0.049$), color 4 (S. Yellow) ($p = 0.007$), color 6 (S. Violet) ($p = 0.0016$), color 14 (Yellow) ($p = 0.0068$), color 18 (Red) ($p = 0.00032$) and color 20 (Green) ($p = 0.0042$).

4 Discussion and Conclusions

The main objective of this work is to identify the impact of color on brain waves in color-blind and non-color-blind individuals.

Blue amplitudes value is statistically different considering the two samples. The blue amplitude is higher in color-blind individuals, meaning that it causes greater stimulation in brain activity. On the other hand, although the amplitude of the red color is high, it is similar in both samples. Yellow, Blue, and Saturated Green present discrepant values of P100 wave amplitude in both samples. Amplitudes are much higher in non-color-blind individuals than in color-blind individuals. According to this study, different colors can cause different amplitudes and latency in P100 waves, considering color-blind and non-color-blind people. In summary, this study is useful to find metrics to classify colors helping in marketing strategies and design typography. The P100 wave used in this kind of strategies is a novelty in the literature and can be considered for future studies.

The results also suggest that there are differences in the brain responses to other colors. For example, S. Red, Blue, Yellow, S. Yellow and Green amplitudes are higher in non-color-blind individuals, while S. Blue amplitude is higher in color-blind individuals.

To further analyze this difference, the study should be conducted in a larger sample size, and a t-test should be applied to compare the amplitude of colors between color-blind and non-color-blind individuals. This test would enable us to determine whether the difference between the amplitude of colors is significant or not. Additionally, a multiple

linear regression analysis could be applied in order to evaluate the effect of different factors (such as age, gender, etc.) on the amplitude of colors.

Overall, the results of this study suggest that color has a significant impact on brain activity and suggest that color-blind individuals have different responses to color than non-color-blind individuals. This has implications for the design of visual stimuli and may be used to tailor visual stimulation to better meet the needs of specific individuals.

References

1. Rakshit, A., Lahiri, R.: Discriminating different color from EEG signals using interval-type 2 fuzzy space classifier (a neuro-marketing study on the effect of color to cognitive state). In: 1st IEEE International Conference on Power Electronics, Intelligent Control and Energy Systems, ICPEICES 2016, February 201. <https://doi.org/10.1109/ICPEICES.2016.7853388>
2. Clarke, T., Costall, A.: The emotional connotations of color: a qualitative investigation. *Color Res. Appl.* **33**(5), 406–410 (2008). <https://doi.org/10.1002/COL.20435>
3. Chai, M.T., et al.: Exploring EEG effective connectivity network in estimating influence of color on emotion and memory. *Front. Neuroinform.* **13**, 66 (2019). <https://doi.org/10.3389/FNINF.2019.00066/BIBTEX>
4. Haider Ali, S.R.K., Khan, Y., Shah, M.: Color psychology in marketing. *J. Bus. Tour.* **4**(1), 183–190 (2021). <https://doi.org/10.34260/JBT.V4I1.99>
5. Yoto, A., Katsuura, T., Iwanaga, K., Shimomura, Y.: Effects of object color stimuli on human brain activities in perception and attention referred to EEG alpha band response. *J. Physiol. Anthropol.* **26**(3), 373–379 (2007). <https://doi.org/10.2114/JPA2.26.373>
6. Wilms, L., Oberfeld, D.: Color and emotion: effects of hue, saturation, and brightness. *Psychol. Res.* **82**(5), 896–914 (2017). <https://doi.org/10.1007/s00426-017-0880-8>
7. Ko, Y.H., Shen, I.H., Lee, D.S.: Color combinations of visual display terminal (VDT) icon on user preferences and EEG response. *Percept. Mot. Skills* **110**(2), 411–428 (2010). <https://doi.org/10.2466/PMS.110.2.411-428>
8. Wang, Y., Wang, S., Xu, M.: The function of color and structure based on EEG features in landscape recognition. *Int. J. Environ. Res. Public Health* **18**(9), 4866 (2021). <https://doi.org/10.3390/IJERPH18094866>
9. Che, X., Zheng, Y., Chen, X., Song, S., Li, S.: Decoding color visual working memory from EEG signals using graph convolutional neural networks. *Int. J. Neural Syst.* **32**(2), 2250003 (2022). <https://doi.org/10.1142/S0129065722500034>
10. Al-Nafjan, A.: Feature selection of EEG signals in neuromarketing. *PeerJ Comput. Sci.* **8**, e944 (2022). <https://doi.org/10.7717/PEERJ-CS.944>
11. Krigolson, O.E., Williams, C.C., Norton, A., Hassall, C.D., Colino, F.L.: Choosing MUSE: validation of a low-cost, portable EEG system for ERP research. *Front. Neurosci.* **11**(Mar), 109 (2017). <https://doi.org/10.3389/FNINS.2017.00109/BIBTEX>
12. D'Andrade, R.G., Romney, A.K.: A quantitative model for transforming reflectance spectra into the Munsell color space using cone sensitivity functions and opponent process weights. *Proc. Natl. Acad. Sci. U S A* **100**(10), 6281–6286 (2003). <https://doi.org/10.1073/PNAS.1031827100/ASSET/19338334-3275-4800-A5F5-BC9D084FB470/ASSETS/GRAPHIC/PQ1031827006.JPEG>